

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Branch:- CSE

Semester – Second

Section- CSE-1/ CSE-2

Mid-Term Examination March – 2026

Subject Name/ Code: - CSE-2004 (Data Structures and Algorithms)

Max. Marks: 20

Time Allotted: 90 min

- NOTE: 1. All questions are compulsory.
2. Mark answer numbers clearly.

S. No	Question	Marks
1.	(a) For the given C like code, form the recurrence relation and find the Big Oh Complexity: void fun(int n){ if(n<=1) return; fun(n-1); fun(n/2); }	2.5
	(b) Analyze the following pseudocode and determine the tight asymptotic complexity in Θ (Theta) notation- for(i = 1; i <= n; i = i * 2) { for(j = i; j <= n; j = j + i) { for(k = 1; k <= j; k = k * 2) { stmt; } } } }	2.5



2.		Convert the following expression: $A + B*(C^D - E)^{(F + G*H)} - I$ (a) Infix to Postfix (b) Infix to Prefix Using the Stack method with clear and detailed steps.	2.5 2.5
3.		For Tower of Hanoi with 5 disks: a) Write recurrence relation b) Find total moves	2.5 2.5
4.	(a)	If a doubly linked list has n nodes, derive the minimum number of pointer updates required to delete the middle node.	2.5
	(b)	Write the pseudocode to create a doubly linked list and then delete every node that has a greater value node on its right side.	2.5